

8.1 Introduction to Tensors

Multi-way arrays and why flattening loses structure.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow *Introduction to Tensor Decompositions and their Applications in Machine Learning*.

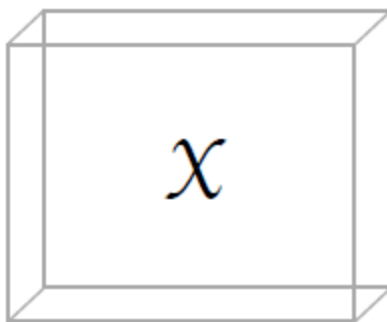
1. Brief history

Tensors generalize matrices to higher dimensions: multidimensional arrays.

They and their decompositions appear in 1927, then sit unused in computer science until late in the 20th century. Cheaper compute and better multilinear algebra — especially in the last decade — moved them into statistics, data science, and machine learning.

2. Tensor basics

A tensor is a multi-way collection of numbers, usually from $\mathbb{R}$. The simplest high-dimensional case is a three-dimensional array: a **data cube**. The same notation extends to more modes.

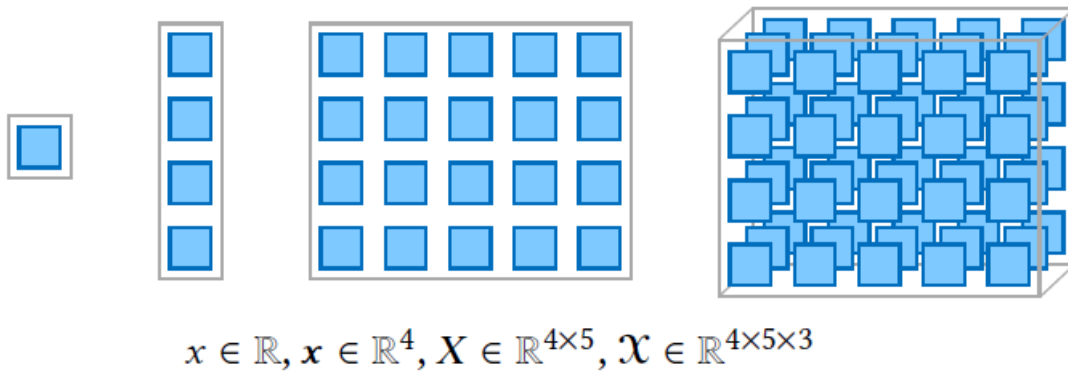


3. Tensor order

The **order** of a tensor is the number of dimensions.

Object	Order	Notation
Scalar	0	$x \in \mathbb{R}$
Vector	1	$\mathbf{x} \in \mathbb{R}^{I_1}$
Matrix	2	$\mathbf{X} \in \mathbb{R}^{I_1 \times I_2}$
Higher-order tensor	N	$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$

Each I is the length of that mode.



4. Tensor indexing

Fix some indices to get a subarray.

- **Fibers:** fix every index but one.
- **Slices** (slabs): fix every index but two.

For a third-order tensor, fibers are $x_{:jk} = x_{jk}$ (column), $x_{i:k}$ (row), $x_{ij:}$ (tube). Slices are $\mathcal{X}_{::k} = \mathcal{X}_k$ (frontal), $\mathcal{X}_{:j:}$ (lateral), $\mathcal{X}_{i::}$ (horizontal).



Column, row, and tube fibers of a mode-3 tensor

5. Outer and inner product

The vector **outer product** is the product of the entries, written $\odot$. For two length- n vectors **a** and **b**,

$$\mathbf{X} = \mathbf{a} \odot \mathbf{b} = \mathbf{a}\mathbf{b}^\top.$$

For N vectors the same product is a tensor:

$$\mathcal{X} = \mathbf{a}^{(1)} \odot \mathbf{a}^{(2)} \odot \dots \odot \mathbf{a}^{(N)},$$

$$\text{with } x_{i_1 i_2 \dots i_N} = a_{i_1}^{(1)} a_{i_2}^{(2)} \dots a_{i_N}^{(N)}.$$

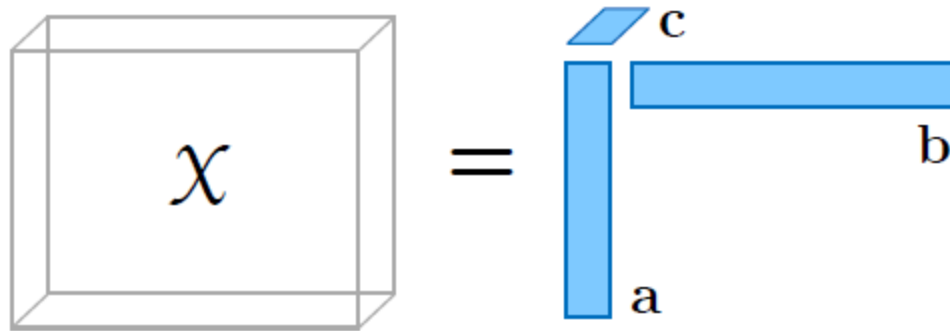
The **inner product** of two length- n vectors is a scalar:

$$x = \langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b} = \sum_{i=1}^n a_i b_i.$$

6. Rank-1 tensors

An N -way tensor is **rank-1** if it is exactly the outer product of N vectors. Adding a mode is a new scaling of a sub-tensor.

A rank-1 matrix: $\mathbf{X} = \mathbf{a} \odot \mathbf{b}$. A rank-1 3-way tensor: $\mathcal{X} = \mathbf{a} \odot \mathbf{b} \odot \mathbf{c}$. The N -way form is the product above.



A rank-1 mode-3 tensor

7. Tensor rank

The **rank** $\text{rank}(\mathcal{X}) = R$ is the smallest number of rank-1 tensors that sum to $\mathcal{X}$:

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{a}_r^{(1)} \odot \mathbf{a}_r^{(2)} \odot \dots \odot \mathbf{a}_r^{(N)}.$$

The **factor matrices** $\mathbf{A}$ store those vectors as columns:

$$\mathbf{A} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_R].$$

The weights λ_r (with $\lambda \in \mathbb{R}^R$) often absorb column norms, typically so each column has squared length one.

Practice

1. A matrix is a 2-way array. What is a tensor adding?
2. What is a **fiber**, and what is a **slice**?
3. What does $\text{rank}(\mathcal{X}) = R$ mean in terms of rank-1 tensors?